

Task1 :

```
Apr 23 23:28
affan@affan-VirtualBox: ~
#include <pthread.h>
#include <stdio.h>
#include <stdlib.h>
void *thread_f(void *thread_no) {
    int tno = *((int *)thread_no);
    printf("thread %d\n", tno+1);
    pthread_exit(NULL);
}
int main() {
    pthread_t threads[3];
    int thread_ids[3];
    int rc;
    for (int i = 0; i < 3; i++) {
        thread_ids[i] = i;
        rc = pthread_create(&threads[i], NULL, thread_f, &thread_ids[i]);
        if (rc) {
            printf("error %d\n", rc);
            exit(-1);
        }
    }
    for (int i = 0; i < 3; i++){
        pthread_join(threads[i], NULL);
        printf("thread %d returns 0\n", i+1);
    }
    return 0;
}
affan@affan-VirtualBox:~$ ./task1.c
bash: ./task1.c: Permission denied
affan@affan-VirtualBox:~$ ./task1
thread 1
thread 2
thread 3
thread 1 returns 0
thread 2 returns 0
thread 3 returns 0
affan@affan-VirtualBox:~$ S
```

Task2:

```
Apr 23 23:46
affan@affan-VirtualBox: ~
affan@affan-VirtualBox:~$ gcc task2.c -o task2 -lpthread
affan@affan-VirtualBox:~$ cat task2.c
#include <pthread.h>
#include <stdio.h>
#include <stdlib.h>
void *thread_f(void *nums) {
    int num = *((int *)nums);
    printf("table of %d: ", num);
    for (int i = 1; i < num; i++) {
        printf(" %d", num*i);
    }
    printf("\n");
    pthread_exit(NULL);
}
int main() {
    pthread_t threads[4];
    int nums[] = {5, 6, 7, 8};
    int rc;
    for (int i = 0; i < 4; i++) {
        rc = pthread_create(&threads[i], NULL, thread_f, &nums[i]);
        if (rc) {
            printf("error %d\n", rc);
            exit(-1);
        }
    }
    for (int i = 0; i < 4; i++){
        pthread_join(threads[i], NULL);
    }
    return 0;
}
affan@affan-VirtualBox:~$ ./task2
table of 5: 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100 105 110 115 120 125 130 135 140 145 150 155 160 165 170 175 180 185 190 195 200 205 210 215 220 225 230 235 240 245 250 2
55 260 265 270 275 280 285 290 295 300 305 310 315 320 325 330 335 340 345 350 355 360 365 370 375 380 385 390 395 400 405 410 415 420 425 430 435 440 445 450 455 460 465 470 475 480 485 490 495
500 505 510 515 520 525 530 535 540 545 550 555 560 565 570 575 580 585 590 595 600 605 610 615 620 625 630 635 640 645 650 655 660 665 670 675 680 685 690 695 700 705 710 715 720 725 730 735 7
40 745 750 755 760 765 770 775 780 785 790 795 800 805 810 815 820 825 830 835 840 845 850 855 860 865 870 875 880 885 890 895 900 905 910 915 920 925 930 935 940 945 950 955 960 965 970 975 980
985 990 995
table of 6: 6 12 18 24 30 36 42 48 54 60 66 72 78 84 90 96 102 108 114 120 126 132 138 144 150 156 162 168 174 180 186 192 198 204 210 216 222 228 234 240 246 252 258 264 270 276 282 288 294 30
0 306 312 318 324 330 336 342 348 354 360 366 372 378 384 390 396 402 408 414 420 426 432 438 444 450 456 462 468 474 480 486 492 498 504 510 516 522 528 534 540 546 552 558 564 570 576 582 588
594 600 606 612 618 624 630 636 642 648 654 660 666 672 678 684 690 696 702 708 714 720 726 732 738 744 750 756 762 768 774 780 786 792 798 804 810 816 822 828 834 840 846 852 858 864 870 876 88
2 888 894 900 906 912 918 924 930 936 942 948 954 960 966 972 978 984 990 996
table of 7: 7 14 21 28 35 42 49 56 63 70 77 84 91 98 105 112 119 126 133 140 147 154 161 168 175 182 189 196 203 210 217 224 231 238 245 252 259 266 273 280 287 294 301 308 315 322 329 336 343
350 357 364 371 378 385 392 399 406 413 420 427 434 441 448 455 462 469 476 483 490 497 504 511 518 525 532 539 546 553 560 567 574 581 588 595 602 609 616 623 630 637 644 651 658 665 672 679 68
6 693 700 707 714 721 728 735 742 749 756 763 770 777 784 791 798 805 812 819 826 833 840 847 854 861 868 875 882 889 896 903 910 917 924 931 938 945 952 959 966 973 980 987 994
table of 8: 8 16 24 32 40 48 56 64 72 80 88 96 104 112 120 128 136 144 152 160 168 176 184 192 200 208 216 224 232 240 248 256 264 272 280 288 296 304 312 320 328 336 344 352 360 368 376 384 39
2 400 408 416 424 432 440 448 456 464 472 480 488 496 504 512 520 528 536 544 552 560 568 576 584 592 600 608 616 624 632 640 648 656 664 672 680 688 696 704 712 720 728 736 744 752 760 768 776
784 792 800 808 816 824 832 840 848 856 864 872 880 888 896 904 912 920 928 936 944 952 960 968 976 984 992
affan@affan-VirtualBox:~$
```

Task3:

```
affan@affan-VirtualBox: ~  
affan@affan-VirtualBox:~$ cat task3.c  
#include <stdio.h>  
#include <stdlib.h>  
#include <pthread.h>  
#include <unistd.h>  
  
typedef struct {  
    int thread_id;  
    int iteration;  
} thread_data;  
void* thread_func(void* arg) {  
    thread_data* data = (thread_data*)arg;  
    printf("Hello from thread %d - I was created in iteration %d\n",  
        data->thread_id, data->iteration);  
    free(data);  
    return NULL;  
}  
  
int main(int argc, char* argv[]) {  
    int N = atoi(argv[1]);  
    pthread_t threads[N];  
    int batch_size = 4;  
    int iteration = 0;  
  
    for (int i = 0; i < N; i++) {  
        if (i % batch_size == 0) {  
            printf("Main thread: Creating batch at iteration %d\n", iteration);  
            sleep(1);  
            iteration++;  
        }  
  
        thread_data* data = malloc(sizeof(thread_data));  
        data->thread_id = i + 1;  
        data->iteration = iteration;  
  
        pthread_create(&threads[i], NULL, thread_func, data);  
    }  
  
    for (int i = 0; i < N; i++) {  
        pthread_join(threads[i], NULL);  
    }  
  
    return 0;  
}
```

```
affan@affan-VirtualBox: ~  
affan@affan-VirtualBox:~$ gcc task3.c -o task3 -lpthread  
affan@affan-VirtualBox:~$ ./task3 12  
Main thread: Creating batch at iteration 0  
Main thread: Creating batch at iteration 1  
Hello from thread 1 - I was created in iteration 1  
Hello from thread 2 - I was created in iteration 1  
Hello from thread 3 - I was created in iteration 1  
Hello from thread 4 - I was created in iteration 1  
Main thread: Creating batch at iteration 2  
Hello from thread 5 - I was created in iteration 2  
Hello from thread 6 - I was created in iteration 2  
Hello from thread 7 - I was created in iteration 2  
Hello from thread 8 - I was created in iteration 2  
Hello from thread 9 - I was created in iteration 3  
Hello from thread 10 - I was created in iteration 3  
Hello from thread 11 - I was created in iteration 3  
Hello from thread 12 - I was created in iteration 3  
affan@affan-VirtualBox:~$
```

Task4:

```
affan@affan-VirtualBox: ~  
#include <pthread.h>  
#include <stdlib.h>  
#include <stdio.h>  
#define TOTAL_THREADS 5  
#define ARRAY_SIZE 10  
int arr[ARRAY_SIZE];  
void *sum_Array(void *arg){  
    int tid = *((int *)arg);  
    int start = tid*(ARRAY_SIZE/TOTAL_THREADS);  
    int end = start + (ARRAY_SIZE/TOTAL_THREADS);  
    int sum=0;  
    for(int i =start;i<end;i++){  
        sum+=arr[i];  
    }  
    printf("thread %d calculated partial sum %d\n",tid,sum);  
    return (void*)(long)sum;  
}  
int main(){  
    pthread_t threads[TOTAL_THREADS];  
    int thread_args[TOTAL_THREADS];  
    void *thread_results[TOTAL_THREADS];  
    long total_sum=0;  
    for(int i =0 ;i<ARRAY_SIZE;i++){  
        arr[i]=rand()%1000;  
        printf("element [%d] of arr: %d\n",i+1,arr[i]);  
    }  
    for(int i =0 ;i<TOTAL_THREADS;i++){  
        thread_args[i]=i;  
        pthread_create(&threads[i],NULL,sum_Array,(void*)&thread_args[i]);  
    }  
    for(int i =0 ;i<TOTAL_THREADS;i++){  
        pthread_join(threads[i], &thread_results[i]);  
        total_sum +=(long)thread_results[i];  
    }  
    printf("total sum: %ld\n",total_sum);  
    return 0;  
}  
affan@affan-VirtualBox:~$  
affan@affan-VirtualBox:~$ cat task4.c  
affan@affan-VirtualBox:~$ gcc task4.c -o task4 -lpthread  
affan@affan-VirtualBox:~$ ./task4  
element [1] of arr: 383  
element [2] of arr: 886  
element [3] of arr: 777  
element [4] of arr: 915  
element [5] of arr: 793  
element [6] of arr: 335  
element [7] of arr: 386  
element [8] of arr: 492  
element [9] of arr: 649  
element [10] of arr: 421  
thread 0 calculated partial sum 1269  
thread 1 calculated partial sum 1692  
thread 2 calculated partial sum 1128  
thread 3 calculated partial sum 878  
thread 4 calculated partial sum 1070  
total sum: 6037  
affan@affan-VirtualBox:~$
```

task5:

```
affan@affan-VirtualBox: ~  
Volunteer 2: Announcement for participant 85  
Volunteer 4: Resolved query for participant 75  
Volunteer 3: Sponsor info for participant 95  
Volunteer 2: Announcement for participant 86  
Volunteer 4: Resolved query for participant 76  
Volunteer 2: Announcement for participant 87  
Volunteer 3: Sponsor info for participant 96  
Volunteer 4: Resolved query for participant 77  
Volunteer 3: Sponsor info for participant 97  
Volunteer 2: Announcement for participant 88  
Volunteer 3: Sponsor info for participant 98  
Volunteer 4: Resolved query for participant 78  
Volunteer 2: Announcement for participant 89  
Volunteer 3: Sponsor info for participant 99  
Volunteer 4: Resolved query for participant 79  
Volunteer 2: Announcement for participant 90  
Volunteer 3: Sponsor info for participant 100  
Volunteer 4: Resolved query for participant 80  
Volunteer 2: Announcement for participant 91  
Volunteer 2: Announcement for participant 92  
Volunteer 4: Resolved query for participant 81  
Volunteer 2: Announcement for participant 93  
Volunteer 4: Resolved query for participant 82  
Volunteer 2: Announcement for participant 94  
Volunteer 4: Resolved query for participant 83  
Volunteer 2: Announcement for participant 95  
Volunteer 2: Announcement for participant 96  
Volunteer 4: Resolved query for participant 84  
Volunteer 2: Announcement for participant 97  
Volunteer 4: Resolved query for participant 85  
Volunteer 2: Announcement for participant 98  
Volunteer 4: Resolved query for participant 86  
Volunteer 2: Announcement for participant 99  
Volunteer 4: Resolved query for participant 87  
Volunteer 2: Announcement for participant 100  
Volunteer 4: Resolved query for participant 88  
Volunteer 4: Resolved query for participant 89  
Volunteer 4: Resolved query for participant 90  
Volunteer 4: Resolved query for participant 91  
Volunteer 4: Resolved query for participant 92  
Volunteer 4: Resolved query for participant 93  
Volunteer 4: Resolved query for participant 94  
Volunteer 4: Resolved query for participant 95  
Volunteer 4: Resolved query for participant 96  
Volunteer 4: Resolved query for participant 97  
Volunteer 4: Resolved query for participant 98  
Volunteer 4: Resolved query for participant 99  
Volunteer 4: Resolved query for participant 100  
All tasks completed.  
affan@affan-VirtualBox:~$
```

```
Apr 23 23:59
affan@affan-VirtualBox: ~
affan@affan-VirtualBox:~$ cat tasks.c
#include <stdio.h>
#include <pthread.h>
#include <unistd.h>
#define PARTICIPANTS 100

void* registration(void* arg) {
    for (int i = 1; i <= PARTICIPANTS; i++) {
        printf("Volunteer 1: Registered participant %d\n", i);
        usleep(5000);
    }
    return NULL;
}

void* announcements(void* arg) {
    for (int i = 1; i <= PARTICIPANTS; i++) {
        printf("Volunteer 2: Announcement for participant %d\n", i);
        usleep(7000);
    }
    return NULL;
}

void* sponsors(void* arg) {
    for (int i = 1; i <= PARTICIPANTS; i++) {
        printf("Volunteer 3: Sponsor info for participant %d\n", i);
        usleep(6000);
    }
    return NULL;
}

void* queries(void* arg) {
    for (int i = 1; i <= PARTICIPANTS; i++) {
        printf("Volunteer 4: Resolved query for participant %d\n", i);
        usleep(8000);
    }
    return NULL;
}

int main() {
    pthread_t v1, v2, v3, v4;

    pthread_create(&v1, NULL, registration, NULL);
    pthread_create(&v2, NULL, announcements, NULL);
    pthread_create(&v3, NULL, sponsors, NULL);
    pthread_create(&v4, NULL, queries, NULL);

    pthread_join(v1, NULL);
    pthread_join(v2, NULL);
    pthread_join(v3, NULL);
    pthread_join(v4, NULL);

    printf("All tasks completed.\n");
    return 0;
}
affan@affan-VirtualBox:~$
```